

Lab 3 Architectural Justification

Local Sports Tournament and Bracket Manager

Problem Statement 54

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Semester - 5

Section - I

Architecture Selection

We chose Layered Architecture for the Local Sports Tournament and Bracket Manager. The browser presentation layer calls an application entry layer for authentication, authorization, and routing. Business components manage registration, brackets, scores, standings and publication; repositories isolate database access. These modules run in one server application with one transactional database.

Architectural Style Analysis

Style	Suitability for this scenario
Layered	Selected: separates tournament rules from screens and storage while keeping related updates in one transaction.
Microservices	Independent scaling is possible, but distributed updates and extra network calls complicate score, bracket and standings consistency.
Client Server	Fits browser access and centralized control, but alone does not define internal modules. It describes deployment here; layers define the server design.

Two Scenario Specific Reasons

1. Separation of concerns: registration validation, seeded single or double elimination, and configured tiebreakers have distinct responsibilities. Their interfaces allow rules to change without rewriting the browser or database code.
2. Consistent results: confirming a score, advancing teams, recalculating standings and recording the audit entry share one transaction. A duplicate result or invalid next slot rejects the update; failed writes roll back together. Publication creates a versioned snapshot so viewers see the same approved results.

Security Advantage

The entry layer checks every protected request on the server. Only an authenticated Tournament Director may create tournaments, confirm scores, generate brackets or publish results. Captains may register their own teams; public users have published read access. Repositories are not exposed to browsers. Successful Director writes include the user, action and timestamp in the same transaction; HTTPS protects client traffic.

Performance Benefit

In process business calls avoid service to service network latency. Indexed match and team access, affected team recalculation, and stored publication snapshots reduce repeated work. The design targets the SRS requirement that at least 95% of 1,000 valid score submissions recalculate within 200 ms with 100 concurrent users. This is a design target and must be verified by implementation load tests.